Project Proposal: Tokenizers and Models for Human Promoter Detection

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Problem and motivation

We will classify 300-base-pair human DNA sequences as promoter or non-promoter, and measure how tokenizer and model choice affect predictive power. Promoters are the regions just upstream of a gene where transcription begins.

Applications

Most human promoters are already mapped by experiments, so the value of a promoter model lies in what it enables beyond the reference genome:

- **Annotating new genomes.** The Earth BioGenome Project aims to sequence all of the roughly 1.8 million named eukaryotic species (Lewin et al., 2018). Lab assays that map promoters cannot keep pace with that, so models trained on well-studied genomes give a fast first-pass map of where genes start.
- **Interpreting disease mutations.** Around 90% of disease-associated variants found in genome-wide association studies lie outside protein-coding genes, many in regulatory DNA (Maurano et al., 2012). A promoter model can score a mutation by comparing its prediction on the normal and mutated sequence, which helps rank variants for follow-up. Mutations in the *TERT* gene promoter, for example, are found in most melanomas and glioblastomas and switch on a gene that lets tumor cells keep dividing (Huang et al., 2013).
- **Designing gene therapies.** Viral vectors used in gene therapy carry only about 4.7 kb of DNA, so they need short, reliable promoters to drive the therapeutic gene (Naso et al., 2017). A classifier can screen candidate synthetic promoters before costly lab testing.
- **Clinical and diagnostic sequencing.** As whole-genome sequencing becomes routine in clinics, most of a patient's variants fall in non-coding DNA that current reports largely ignore. Fast, accurate regulatory models are a step toward flagging the ones that matter.
- **Learning the grammar of gene regulation.** Interpretability methods show which sequence patterns a model relies on. These can confirm known motifs or suggest new candidates for biologists to test.

Our project contributes to these goals by measuring which tokenizers and model types capture promoter signals best, and at what computational cost. Those findings carry over directly to the harder applied problems above.

We chose this problem because DNA is a language with no natural word boundaries. That makes tokenization (single nucleotides, k-mers, or learned subwords) a real experimental variable, not a preprocessing detail. Earlier DNA language models have used overlapping k-mers (Ji et al., 2021) and learned BPE subwords (Zhou et al., 2024). The task uses lab-validated labels and has published benchmark scores, so our results can be judged against prior work.

Dataset

We will use the `prom_300_all` task from the Genome Understanding Evaluation (GUE) benchmark released with DNABERT-2 (Zhou et al., 2024). In the GUE data set, positive examples are human promoters from EPDnew, a database of experimentally validated promoters (Dreos et al., 2013). Negatives are non-promoter human sequences of the same length.

- Approximately 47,000 training, 5,900 validation, and 5,900 test sequences
- Each sequence is 300 bp, roughly balanced between the two classes
- Distributed as CSV files of sequence and label, with fixed train, dev, and test splits

NLP methods

We will build models from four concepts, ranging from classical baselines to a customized pretrained transformer, and pair each with the tokenizers it can use.

Concept	Model	Tokenizers tested
Classical	Logistic regression on k-mer counts and TF-IDF	k-mers, k = 3 to 6
Recurrent network	BiLSTM with learned embeddings	Single nucleotide, overlapping k-mers, BPE
Pretrained transformer	DNABERT-2 (117M), as frozen embeddings with logistic regression and fully fine-tuned	DNABERT-2's BPE

Interpretability	Integrated Gradients on the best RNN and transformer	DNABERT-2's BPE, with token scores mapped back to individual nucleotides

The classical models are standard. The recurrent models and BPE tokenizers we will build ourselves, training BPE vocabularies of two sizes (512 and 4,096 tokens) on the training sequences. DNABERT-2 (Zhou et al., 2024) will be customized with a new classification head and fully fine-tuned. Integrated Gradients (Sundararajan et al., 2017) will show which positions drive the best DNABERT-2 model's predictions; we will check whether attributions averaged across test promoters peak where known promoter elements sit, such as the TATA box about 25 to 30 bp before the transcription start site.

Packages

Package	Use
pandas, NumPy	Loading and preparing sequence data
scikit-learn	k-mer vectorizers, logistic regression, metrics
PyTorch	BiLSTM models and training loops
Hugging Face `transformers`, `tokenizers`	Loading and fine-tuning DNABERT-2, training BPE vocabularies
Captum	Integrated Gradients for PyTorch models
Matplotlib, seaborn	Results figures and attribution plots

We chose these because they are the standard Python tools for each step, they work together (Captum plugs into PyTorch and Hugging Face models directly), and the DNABERT-2 authors publish their model and fine-tuning code for this stack (Zhou et al., 2024).

NLP tasks

- **Sequence classification:** binary promoter detection, the main task
- **Tokenization:** building and comparing character, k-mer, and BPE tokenizers for DNA
- **Representation learning:** comparing learned embeddings (RNNs) with pretrained ones (DNABERT-2)
- **Model interpretation:** attributing predictions to sequence positions and motifs

Evaluating performance

Our primary metric is the Matthews correlation coefficient (MCC), which GUE reports, so our numbers are directly comparable to the published DNABERT-2 score (about 0.86 on this task; Zhou et al., 2024). We will also report accuracy, F1, and ROC-AUC on the held-out test set, along with confusion matrices for the best models.

All hyperparameter tuning (hyperparameters to be chosen from among: learning rate, embedding size, hidden size, k, BPE vocabulary size) will use the validation split only. We will use early stopping on validation loss to limit overfitting, and we will touch the test set once per final model. We will also compare training time and parameter count, since a small model that nearly matches DNABERT-2 is a useful result in itself.

Schedule

- **Week 1 (Oct 26 to Nov 1):** Submit proposal; set up Github; set up the Python and GPU environment; load DNABERT-2
- **Week 2 (Nov 2 to 8):** Download data and run exploratory analysis; write shared evaluation script; build k-mer tokenizers; train logistic regression baselines
- **Week 3 (Nov 9 to 15):** Build BPE tokenizers; train BiLSTM models across tokenizers; run DNABERT-2 frozen-embedding baseline
- **Week 4 (Nov 16 to 22):** Full fine-tuning of DNABERT-2; tune hyperparameters on the validation set
- **Week 5 (Nov 23 to 29):** Final runs on the test set; Integrated Gradients analysis of the best DNABERT-2 model
- **Week 6 (Nov 30 to Dec 6):** Results figures and tables; group report, slides, and repo READMEs; individual reports

References

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Zhou, Z., et al. (2024). DNABERT-2: Efficient foundation model and benchmark for multi-species genome. *ICLR 2024*. https://doi.org/10.48550/arXiv.2306.15006. Code and GUE data: https://github.com/MAGICS-LAB/DNABERT_2